



Now we can "do" the integral.

$$I = \lim_{\substack{n \rightarrow \infty \\ \Delta x \rightarrow 0}} \sum_{i=1}^n f(x_i^*) \Delta x_i$$

$\Delta x = \max_{i=1}^n \Delta x_i$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left[ \left( 5 \left( -2 + \frac{6i-3}{2n} \right) - 2 \right) \frac{3}{n} \right]$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left( -12 + \frac{5(6i-3)}{2n} \right) \frac{3}{n}$$

$$= \lim_{n \rightarrow \infty} \sum_{i=1}^n \left( \frac{-36}{n} + \frac{15(6i-3)}{2n^2} \right)$$

$$= \lim_{n \rightarrow \infty} \left[ \left( \sum_{i=1}^n \frac{-36}{n} \right) + \left( \sum_{i=1}^n \frac{15(6i-3)}{2n^2} \right) \right]$$

$$= \lim_{n \rightarrow \infty} \left[ -36 + \frac{1}{2n^2} \sum_{i=1}^n (90i - 45) \right]$$

$$= \lim_{n \rightarrow \infty} \left[ -36 + \frac{90}{2n^2} \left( \sum_{i=1}^n i \right) - \frac{45}{2n} \right]$$

$$= \lim_{n \rightarrow \infty} \left[ -36 + \frac{90}{2n^2} \frac{n(n+1)}{2} - \frac{45}{2n} \right]$$

$$= \lim_{n \rightarrow \infty} \left[ -36 + \frac{90}{4} + \frac{90}{4n} - \frac{45}{2n} \right]$$



But with rep. points ~~not~~  $\sum_{i=1}^n f(x_i^*) \Delta x_i$

$$x_i^* = -2 + (i-1) \frac{3}{n}$$

$$I = \lim_{n \rightarrow \infty} \sum_{i=1}^n \textcircled{3} \left( -2 + (i-1) \frac{3}{n} \right)^2 \frac{\textcircled{3}}{n}$$

$$= 9 \lim_{n \rightarrow \infty} \left( \frac{1}{n} \sum_{i=1}^n \left( \overset{(a+b)^2}{-2 + (i-1) \frac{3}{n}} \right)^2 \right)$$

$$= 9 \lim_{n \rightarrow \infty} \left( \frac{1}{n} \sum_{i=1}^n \left( 4 - 4(i-1) \frac{3}{n} + (i-1)^2 \frac{9}{n^2} \right) \right)$$

$$= 9 \lim_{n \rightarrow \infty} \left( 4 - \frac{12}{n^2} \sum_{i=1}^n (i-1) + \frac{9}{n^3} \sum_{i=1}^n (i-1)^2 \right)$$

$$= 9 \lim_{n \rightarrow \infty} \left( 4 - \frac{12}{n^2} \underbrace{\sum_{j=1}^{n-1} j} + \frac{9}{n^3} \underbrace{\sum_{j=1}^{n-1} j^2} \right)$$

$$= 9 \lim_{n \rightarrow \infty} \left( 4 - \frac{12}{n^2} \frac{n-1}{2} \cdot n + \frac{9}{n^3} \frac{(n-1)n(2n-1)}{6} \right)$$

$$= 9 \left( 4 - 6 + \frac{9}{3} \right) \checkmark$$

$$= 9 \left( -2 + \frac{9}{3} \right)$$

$$= 9(1) = 9$$

I think.  
Check carefully.

